

# Homework - 3

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1. a. Number of instructions :  $\frac{\text{(Processor Speed)}}{\text{CPI}}$   
per second

So,

$$P1 = (3.0 * 10^9) / (1.5) = 2.0 * 10^9 \text{ instructions per sec}$$

$$P2 = (2.5 * 10^9) / (1.0) = 2.5 * 10^9 \text{ instructions per sec}$$

$$P3 = (4.0 * 10^9) / (2.2) = 1.8 * 10^9 \text{ instructions per sec}$$

P2 has the highest performance.

- b. Number of instructions  $\cdot \left[ \frac{\text{no. of instructions}}{\text{per second}} \right] * \left[ \frac{\text{Execution Time}}{\text{Time}} \right]$

For,

$$P1 = 2.0 * 10^9 * 10 = 2.0 * 10^{10} \text{ instructions}$$

$$P2 = 2.5 * 10^9 * 10 = 2.5 * 10^{10} \text{ instructions}$$

$$P3 = 1.8 * 10^9 * 10 = 1.8 * 10^{10} \text{ instructions}$$

2. Instruction count  $2.389 * 10^{12}$

Execution Time = 750 sec

Reference Time = 9650 sec

- a.  $CPI = \frac{\text{Executine Time}}{(\text{no. of instructions} * \text{clock cycle time})}$

$$\text{clock cycle time} = 0.333 \text{ ns} \\ = 0.333 * 10^{-9} \text{ sec.}$$

$$CPI = \frac{750}{2.389 * 10^{12} * 0.333 * 10^{-9}} = \frac{750}{795.537} = 0.9428 \\ = 0.94$$

b.  $\text{SPEC ratio} = \frac{\text{Reference Time}}{\text{Execution Time}}$

$$= \frac{9650}{750} = 12.867$$

3. P1 clock rate = 4 GHz      P2 clock rate = 3 GHz  
 P1 average CPI = 0.9      P2 average CPI = 0.75  
 P1 instructions =  $5.0 \times 10^9$       P2 instructions =  $1.0 \times 10^9$

a.  $\text{Execution Time} = \frac{\text{no. of instructions} \times \text{CPI}}{\text{clock rate}}$

For P1,

$$\text{Execution Time} = \frac{(5.0 \times 10^9) \times 0.9}{4 \times 10^9} = \frac{4.5}{4} = 1.125 \text{ sec}$$

For P2,

$$\text{Execution Time} = \frac{(1.0 \times 10^9) \times 0.75}{3 \times 10^9} = \frac{0.75}{3} = 0.25 \text{ sec}$$

After calculating execution time, we can see that P2 has a lower execution time which in terms of performance that P2 is faster.

b. P1: MIPS:  $(5.0 \times 10^9 / 1.125) / 10^6 = 4444.44$   
 P2: MIPS:  $(1.0 \times 10^9 / 0.25) / 10^6 = 4000$

Even though the MIPS of P1 is higher, P2 has a lower execution time

#### 4.a Using Amdahl's Law

$$T_{\text{new}} = \frac{T_{\text{affected}}}{n} + T_{\text{unaffected}}$$

$$n = \frac{1}{0.8} = 1.25 \quad \text{as the time reduces 20\%}$$

$$T_{\text{new}} = \frac{70}{1.25} + 180 = 236$$

$$\text{Percentage reduction} = \frac{250 - 236}{250} \times 100$$

$$= \frac{14}{250} \times 100 = 5.6\%$$

b. If only branch time (40 sec) is reduced by 20%.

$$\text{Total time} = \frac{40}{n} + 210 \approx 210$$

$$\text{Percentage reduction} = \frac{250 - 210}{250} \times 100 = 16\%$$

Thus, there will be no benefit by reducing only the time for branch instructions.

5. Instruction count =  $10^6$

For P1:

$$\text{Execution time (A)} = \frac{100000 * 1}{2.5 * 10^9} = 4 * 10^{-5} \text{ sec}$$

$$\text{Execution time (B)} = \frac{200000 * 2}{2.5 * 10^9} = 1.6 * 10^{-4} \text{ sec}$$

$$\text{Execution time (C)} = \frac{500000 * 3}{2.5 * 10^9} = 6 * 10^{-4} \text{ sec}$$

$$\text{Execution time (D)} = \frac{200000 * 3}{2.5 * 10^9} = 2.4 * 10^{-4} \text{ sec}$$

$$\begin{aligned} \text{Overall execution for P1} &= (4 * 10^{-5}) + (1.6 * 10^{-4}) + \\ &\quad (6 * 10^{-4}) + (2.4 * 10^{-4}) \\ &= 1.04 * 10^{-3} \text{ sec} \end{aligned}$$

$$\begin{aligned} \text{Average CPI for P1} &= \frac{(1 * 100000 + 2 * 200000 + 3 * 500000 + 3 * 200000)}{1000000} \\ &= 2.6 \end{aligned}$$

For P2:

$$\text{Execution time (A)} = \frac{100000 * 2}{3 * 10^9} = 6.67 * 10^{-6} \text{ sec}$$

$$\text{Execution time (B)} = \frac{200000 * 2}{3 * 10^9} = 1.33 * 10^{-5} \text{ sec}$$

$$\text{Execution time (C)} = \frac{500000 * 2}{3 * 10^9} = 3.33 * 10^{-5}$$

$$\text{Execution time (D)} = \frac{200000 * 2}{3 * 10^9} = 1.33 * 10^{-5}$$

$$\begin{aligned} \text{Overall execution for P2} &= (6.67 * 10^{-6}) + (1.33 * 10^{-5}) + \\ &\quad (3.33 * 10^{-5}) + (1.33 * 10^{-5}) \\ &= 5.33 * 10^{-5} \end{aligned}$$

$$\begin{aligned} \text{Average CPI for P2} &= \frac{(2 * 100000 + 2 * 200000 + 2 * 500000 \\ &\quad + 2 * 200000)}{1000000} \end{aligned}$$

$$= 2.0$$

- a. P1 has an execution time of  $1.04 * 10^{-3}$  sec; while P2 has an execution time of  $5.33 * 10^{-5}$  sec.  
Therefore, P2 is faster

- b. Average CPI for P1 = 2.6  
Average CPI for P2 = 2.0